



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE : DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	Medical Insurance Cost Predictor	
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Abstract

The **Medical Insurance Cost Predictor** is a Machine Learning-based project developed to predict the estimated medical insurance cost of an individual using different personal, demographic, and health-related factors. Medical insurance charges can vary significantly depending on factors such as **age, gender, Body Mass Index (BMI), number of children, smoking habits, and geographical region**. Understanding the relationship between these factors and insurance costs can help in making better financial and healthcare-related decisions.

In this project, a dataset containing information about individuals and their corresponding medical insurance charges is used to train a **regression-based Machine Learning model**. The collected data is first cleaned and preprocessed to handle inconsistencies and prepare it for model training. Important features are selected, and the dataset is divided into **training and testing sets**. Different regression algorithms, such as **Linear Regression, Decision Tree Regression, and Random Forest Regression**, can be implemented and compared.

The trained model learns the relationship between an individual's characteristics and their medical insurance charges. When details of a new individual are provided, the system uses the learned patterns to **predict the estimated insurance cost**. The performance of the model is measured using evaluation metrics such as **Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² score**.

The proposed system provides a simple and efficient way to estimate medical insurance costs using historical data. It can help users understand the major factors affecting insurance charges and provide an approximate cost estimate. Overall, this project demonstrates the practical application of **Machine Learning, data preprocessing, and regression techniques in the healthcare insurance domain**.